Evaluating Domain Transfer for Martian Terrain Traversability Classification Using Terrestrial Imagery

Mrinank Sivakumar
Big Data Analysis and Project, Master of Artificial Intelligence and Machine Learning
Univerity of Adelaide
Adelaide, Australia
mrinank.sivakumar@student.adelaide.edu.au

***Abstract*—Reliable terrain traversability classification is an integral task for autonomous rovers navigating hazardous extraterrestrial environments. However, the scarcity of labelled Martian image data limits the effectiveness of data-driven approaches. This study investigates whether co-training convolutional neural networks (CNNs) on terrestrial data can enhance traversability classification performance on Martian terrain. Multiple models of increasing complexity (logistic regression, a lightweight Mini-CNN, and a MobileNetV2 encoder) are evaluated using the AI4Mars (Martian) and Freiburg Forest (Earth) datasets. The results indicate that logistic regression and the Mini-CNN improve modestly with Earth co-training, achieving balanced accuracy gains of up to 1.3% and 1.1%, respectively. In contrast, MobileNetV2 achieves the highest overall performance when trained solely on Martian data (94.0% balanced accuracy), but its performance degrades when co-trained with Earth data. These findings suggest that, while domain transfer benefits simpler models, high-capacity models may be more sensitive to domain shift. Based on these results, modern architectures such as MobileNetV2 trained exclusively on domain-relevant data are recommended for efficient onboard deployment, while future work may focus on mitigating cross-domain variance using domain adaptation or style transfer techniques.**

***Index Terms*—Terrain classification, Martian robotics, domain transfer, convolutional neural networks, AI4Mars, Freiburg Forest, co-training, mobile deep learning, autonomous navigation**

I. Introduction

A. Background/Context

Autonomous navigation is a necessary capability for rovers operating in extraterrestrial environments, such as the Moon or Mars. These rovers must identify traversable terrain in real-time to avoid hazards such as steep slopes, soft surfaces, or large obstacles. Traditional rule-based approaches to traversability estimation are often insufficient due to the diversity of planetary surfaces and the unpredictable nature of lighting, shadows, and dust. Data-driven computer vision models, especially those using deep learning, have shown promise in recent years, but are limited by the scarcity of annotated extraterrestrial data.

B. Motivation

Given the high cost and logistical challenges of collecting labelled Martian image data, the field of Martian traversability estimation through vision models is largely unexplored. Despite this fact, deploying non-autonomous or traditional rule-based autonomous rovers could easily end up costing more due to labour, time, and/or accidents. The motivation behind this project was to explore whether the ability of deep learning computer vision models to distinguish between safe and hazardous regions on Mars could be improved in some way. The approach focuses on binary traversability classification at the image patch level using supervised learning methods.

C. Proposed Solution

Thanks to the growing popularity of co-training for robust models, leveraging annotated terrestrial datasets to supplement model training became a plausible option to explore. Although Earth and Mars differ in colour distribution, surface composition, and lighting conditions, they share some fundamental similarities, such as texture patterns and geometric terrain cues.

As such, this study seeks to answer the following question:

> *Can co-training a convolutional neural network (CNN) on both Martian and terrestrial imagery improve binary terrain traversability classification on Mars compared to training on Martian data alone?*

D. Contributions

The main contributions of this study are:

- The development and evaluation of three classification models of varying complexity (logistic regression, Mini-CNN, and MobileNetV2) for binary terrain classification using the AI4Mars and Freiburg Forest datasets.
- A systematic analysis of co-training effects by comparing model performance on Martian test data, with and without terrestrial data augmentation.
- The identification of conditions under which domain transfer improves performance and recommended model configurations for efficient onboard rover deployment.

II. Literature Review

Traversability estimation has been a central challenge in planetary robotics, off-road navigation, and autonomous mobility for years. Past research in this area has covered dataset creation, semantic segmentation, domain adaptation, and real-time learning. This section reviews five studies that were most relevant to this project.

TABLE I
Comparison of Reviewed Literature and This Project

Study	**Approach**	**Self-Supervised**	**Mars Data**	**Key Contribution**
AI4Mars-[19]	Supervised segmentation (DeepLab-V3, SPOC)	No	Yes	High-quality semantic labels for Martian terrain
MTC [1]	Light-weight CNN (Mobile-NetV2)	No	Yes	Cross-mission model generalization for Mars
WVN [7]	DINO-ViT with online memory bank	Yes	No	Real-time traversability learning in natural terrain
AnyTraverse [15]	Vision-language with CLIPSeg	Partial	No	Zero-shot segmentation for off-road terrain
RoadRunner [3]	BEV + sensor fusion (LiDAR + RGB)	Yes	No	Real-time traversability from hindsight self-supervision
This Study	Supervised binary classification (CNNs, RGB only)	No	**Yes**	Evaluates Earth-to-Mars domain transfer with lightweight models

A. AI4MARS: ADataset for Terrain-Aware Autonomous Driving on Mars

Swan et al. [19] created the AI4Mars dataset to support deep learning models in training and validating for Mars terrain classification. It includes more than 16,000 images captured by the Curiosity rover and annotated at the pixel level into five categories: soil, bedrock, sand, big rock, and NULL. The authors evaluated models such as DeepLabv3 and the legacy Soil Property and Object Classification (SPOC) system, and also provided recommendations for model deployment on rovers. This dataset serves as the primary source of Martian imagery in this study.

B. Multi-mission Terrain Classifier for Safe Rover Navigation and Automated Science

Atha et al. [1] proposed a deep learning pipeline for generalizable terrain classification across Mars missions using DeepLabV3+ with a MobileNetV2 encoder. Their models were trained on the AI4MArs dataset, which contained imagery from multiple rovers (Curiosity, Perseverance, Opportunity) and used a unified terrain taxonomy. The study was able to achieve high generalization and emphasize onboard deployability. This paper directly informs our project's use of testing with a light-weight CNN and MobileNetV2 for Martian terrain analysis.

C. Wild Visual Navigation: Fast Traversability Learning via Pre-Trained Models and Online Self-Supervision

Mattamala et al. [7] introduced Wild Visual Navigation (WVN), which is a self-supervised online learning method for traversability estimation. WVN uses DINO-ViT as the foundation for feature extraction and a temporal memory bank to allow the system to learn in real time during deployment. Although this paper is focused on Earth environments, WVN demonstrates the viability of deploying adaptive, self-updating perception systems in natural terrain. The results of this paper suggest that Mars missions using static models may potentially underperform in the future.

D. AnyTraverse: An off-road traversability framework with VLM and human operator in the loop

Sahu et al. [15] presented AnyTraverse, a vision-language traversability framework that combines CLIPSeg for segmentation along with iterative user refinement. Unlike classical CNNs, this method takes advantage of human knowledge/intervention and prompt-driven zero-shot segmentation. AnyTraverse was validated on the RELLIS-3D [6], Freiburg Forest [20], and RUGD datasets. The results it was able to achieve showed strong performance in unfamiliar environments. While this is less applicable to our project, due to the need for prompt tuning, it contrasts with the CNN-based models explored in our study and provides a vision of alternative approaches to domain generalization that could show potential.

E. RoadRunner- Learning Traversability Estimation for Autonomous Off-road Driving

Frey et al. [3] presented RoadRunner, which is a real-time traversability estimation framework tailored for high-speed off-road navigation using a fused camera and LiDAR inputs. Unlike traditional methods that predict traversability based on handcrafted semantic classes and rule-based heuristics, RoadRunner directly predicts traversability and elevation from sensory input. It does so in a fully self-supervised manner using pseudo-ground truths that are generated from hindsight. The model utilizes a Bird's Eye View (BEV) perspective and makes use of a dual-stream architecture. This architecture combines PointPillars and EfficientNet-B0 to handle 3D point cloud and 2D image data, respectively. The RoadRunner system outperforms classical pipelines, such as using X-Racer, in both prediction accuracy and latency (140 ms vs. 500 ms). By achieving robust hazard detection even at long ranges, this study shows the viability and benefits of high-speed learning-based traversability estimation. It also pushes for the further adoption of sensor-fusion models with hindsight supervision, which could prove to be a fruitful future research direction for our project and its unstructured environment.

F. Comparison with Current Project

Table I compares and contrasts the reviewed literature with this research project. Unlike zero-shot or multimodal systems, this project uses a fully supervised, single-modality (RGB) approach. This project explores the effect that domain transfer has from terrestrial to Martian image data on CNN performance by evaluating co-training strategies on lightweight models.

III. Research Methodology

This project follows a structured, five-phase pipeline to evaluate the effectiveness of co-training supervised terrain classification models on Earth and Mars imager data. The methodology is designed to answer whether incorporating terrestrial data (from the Freiburg Forest dataset) improves terrain classification performance on Martian images (from the AI4Mars dataset). Figure 1 illustrates the overall structure of the project.

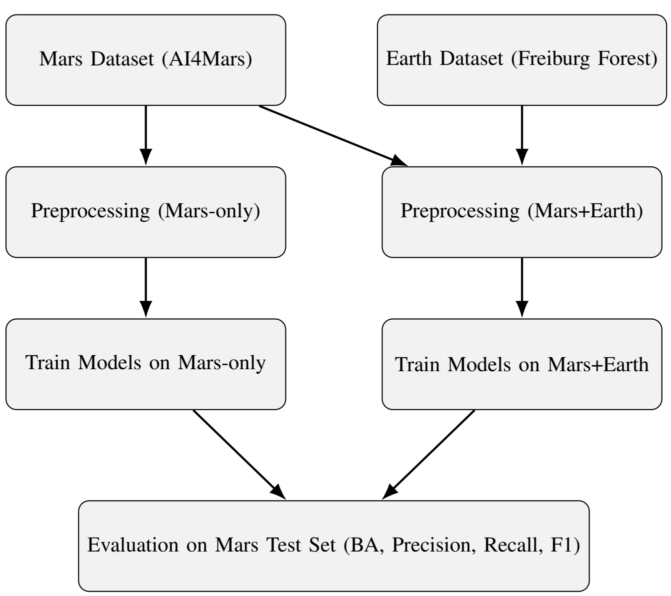

Fig. 1. Methodology overview showing separate model training pipelines for Mars-only and Mars+Earth co-training, followed by common evaluation on the Mars test set.

A. Phase 1: Data Collection and Preparation

Two datasets were used: the AI4Mars dataset [19], which contains pixel-level annotations of Martian terrain captured by the Curiosity rover, and the Freiburg Forest dataset [20], which provides annotated RGB images of terrestrial off-road environments. To ensure consistency, both datasets were preprocessed to extract small image patches (center-cropped to 320×320 pixels) and relabelled into a binary format: 'safe' or 'hazardous' based on the traversability of the terrain.

B. Phase 2: Data Preprocessing and Label Alignment

Due to differences in class labels between datasets, a unified label mapping scheme was applied. In AI4Mars, the 'soil' and 'bedrock' were considered 'safe', while the 'sand', 'big rock' and NULL regions were labelled 'hazardous'. In Freiburg Forest, 'road' was labelled 'safe', while the remaining classes (Grass, Vegetation, Tree, and Obstacle) were labelled 'hazardous'. All the images were flattened to get the mean of each channel, converted to greyscale to compute standard deviation of shades, and had their edge density extracted during feature extraction.

C. Phase 3: Model Selection and Architecture

Three model types were used:

- **Logistic Regression (baseline)** – a simple linear classifier used as a performance reference.
- **Mini-CNN** – a custom lightweight convolutional neural network designed for efficiency.
- **MobileNetV2** – a widely adopted lightweight deep CNN used in embedded and mobile applications.

These models were chosen for their compatibility and history of deployment in mobile applications, making them perfect for rovers operating great distances from earth and human intervention. Each model was trained in two settings: Mars-only and Mars+Earth (co-trained). This design allowed for a controlled comparison of domain transfer performance between the models.

D. Phase 4: Training and Co-Training Setup

The datasets came pre-split into training and testing sets. As such, all of the models were trained using the prepared training images. For testing, Mars-only data was used for all evaluation as it would not make sense to evaluate on terrestrial data. The Mars+Earth setting simply concatenated Earth and Mars training samples prior to model fitting. Binary Cross-Entropy loss was used, models were trained using the Adam optimizer with different batch sizes and epochs per model and trial, and early stopping with a patience of 3 was implemented. The Logistic Regression model was only run once to act as a baseline for performance. Meanwhile, the lightweight CNN was trialed twice: once with a batch size of 32 for 10 epochs and once with a batch size of 64 for 15 epochs. The MobileNet model was also trialed twice: once with a batch size of 16 for 10 epochs and once with a batch size of 32 for 15 epochs.

E. Phase 5: Evaluation and Performance Metrics

The models were evaluated on the Mars test set using the following metrics:

- **Balanced Accuracy (BA)** – to account for class imbalance.

- **Precision, Recall, F1-score** – for both ‘safe’ and ‘hazard’ classes.

Comparisons were made between Mars-only and Mars+Earth training to quantify any performance gain (or drop) due to co-training. The Mini-CNN and MobileNetV2 models were assessed in both 10-epoch and 15-epoch configurations.

IV. Experimental Evaluation

A. Experimental Setup

The experiments were carried out using a binary classification framework for terrain traversability, distinguishing between *safe* and *hazardous* terrain. All of the models were implemented in Python using PyTorch [14], NumPy [9], and scikit-learn [12], with data processing supported by OpenCV [10] and Albumentations [2]. Evaluation was performed on the Mars test set, obtained from the AI4Mars dataset [19]. The input data consisted of RGB images paired with pixel-wise annotated masks.

The two datasets used during training:

- **Mars (AI4Mars)** [19]: high-resolution greyscale images captured by the Curiosity rover and annotated for terrain types.
- **Earth (Freiburg Forest)** [20]: natural forest terrain with RGB inputs and hazard labels derived from semantic segmentation maps.

Each model was trained in two configurations: (1) using Mars data only, and (2) co-training with both Mars and Earth datasets. Three models were evaluated:

- Logistic Regression (baseline)
- Mini-CNN: a lightweight convolutional neural network
- MobileNetV2: a pre-trained convolutional model fine-tuned for this task

To explore performance sensitivity, two sets of hyperparameters were trialed:

1) **10-epoch trial**: Mini-CNN (batch size 32), MobileNetV2 (batch size 16)
2) **15-epoch trial**: Mini-CNN (batch size 64), MobileNetV2 (batch size 32)

Model performance was tracked via:

- Accuracy, Precision, Recall, F1-Score
- Balanced Accuracy (macro average recall)
- Confusion Matrices
- Learning Curves

B. Experimental Results

Figure 2 and Figure 3 show that all models performed better at identifying hazard pixels than safe ones, with co-training (Mars+Earth) occasionally introducing false positives for hazard. MobileNetV2 in particular maintained perfect hazard recall (1.00) across all trials, but this came at the cost of lower safe-class precision in co-training.

Feature projection visualizations using t-SNE (Figure 4) show that Mars and Earth samples form distinct clusters in feature space. This distinction could be an explanation for why the models that were co-trained on Earth samples did not always generalize well to Mars terrain.

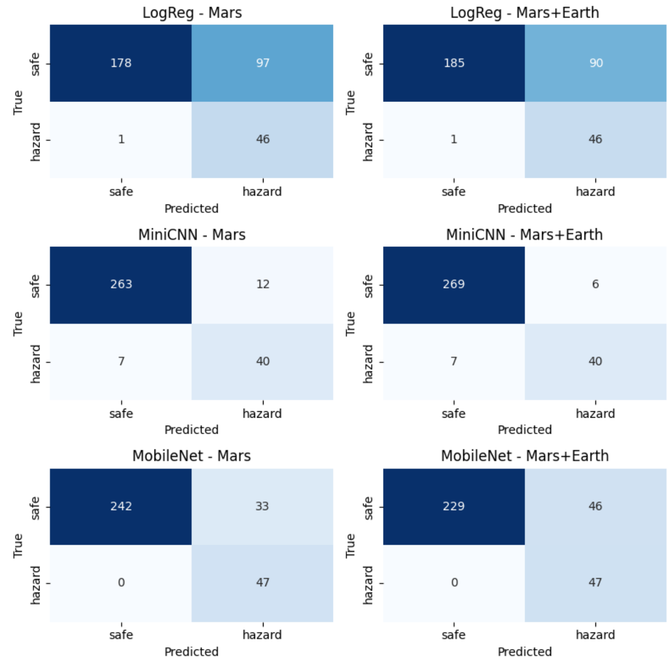

Fig. 2. Confusion matrices for Mars-only vs Mars+Earth across three models (10-epoch trial).

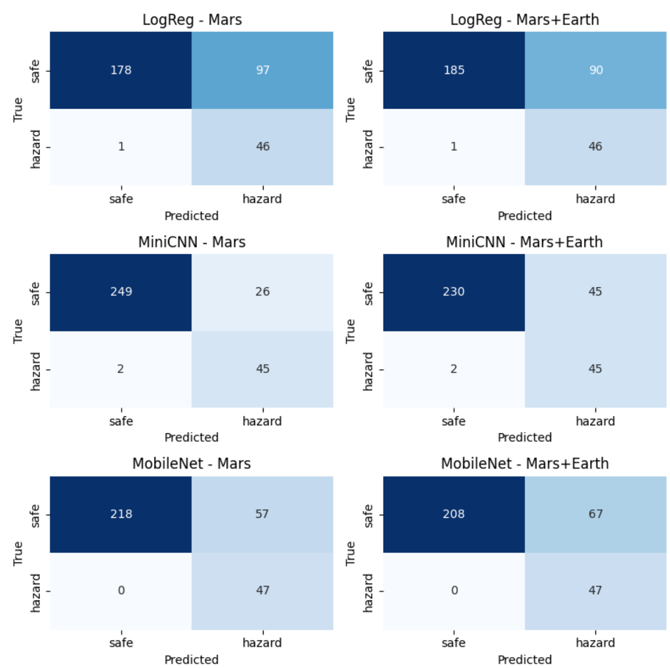

Fig. 3. Confusion matrices for Mars-only vs Mars+Earth across three models (15-epoch trial).

Figure 5 and Figure 6 show that Mars-only training produced higher balanced accuracy in more complex models. For MobileNetV2, adding Earth data actually decreased performance by 2.4% (10-epoch) and 1.8% (15-epoch). This

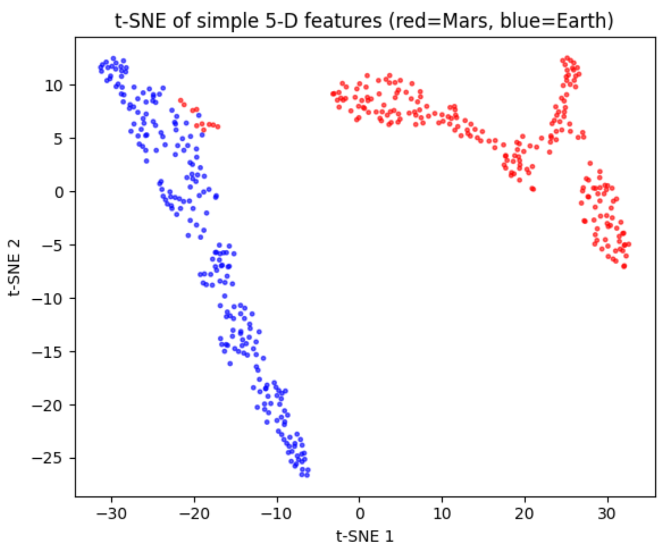

Fig. 4. t-SNE of Mars and Earth 5-D features.

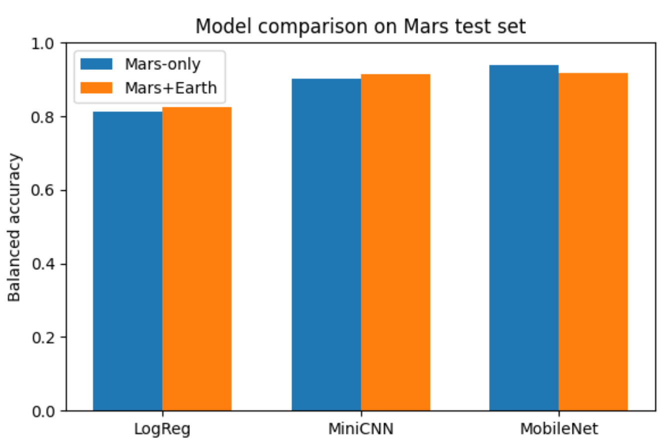

Fig. 5. Balanced accuracy comparison for 10 epoch trial.

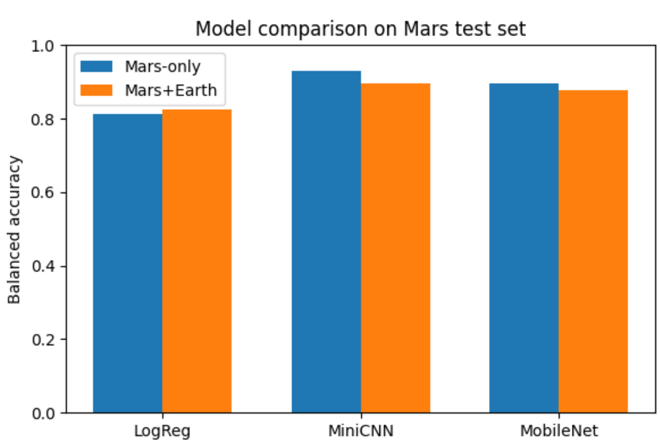

Fig. 6. Balanced accuracy comparison for 15 epoch trial.

suggests that large pre-trained models are already robust to domain shift, and introducing heterogeneous Earth terrain may harm generalization rather than reinforcing it.

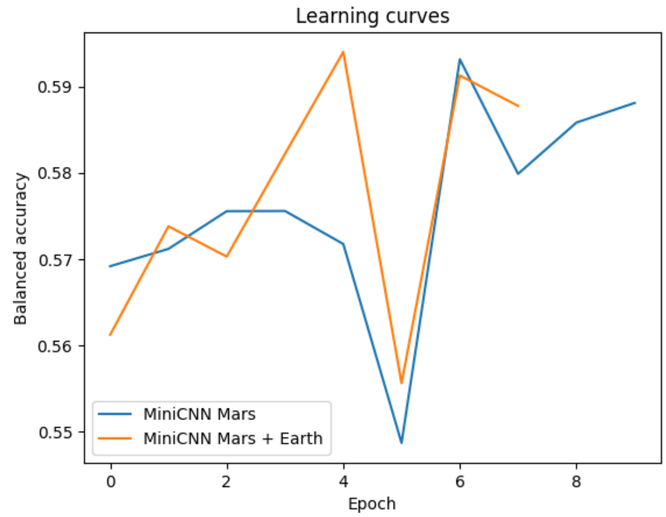

Fig. 7. Mini-CNN learning curves (balanced accuracy) on 10 epoch trial.

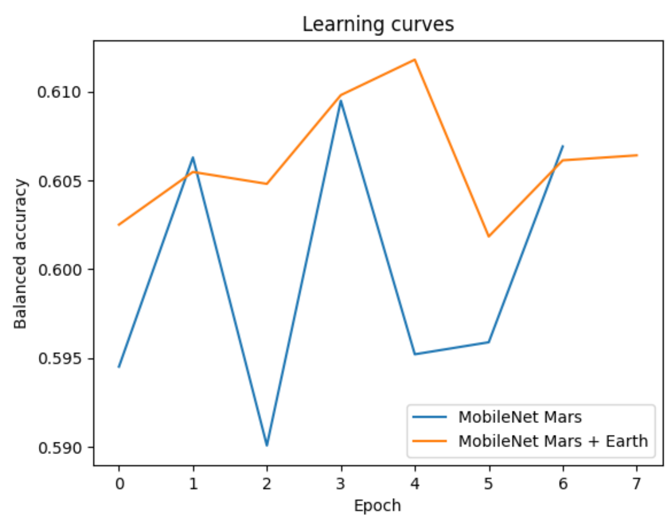

Fig. 8. MobileNetV2 learning curves (balanced accuracy) on 10 epoch trial.

Figures 7, 8, 9, 10 show that Mars+Earth co-training offered minor improvements for the Mini-CNN, especially in early epochs. However, the performance gain plateaued and reversed in the 15-epoch trial, which indicates potential overfitting or dataset incompatibility.

TABLE II
BALANCED ACCURACY ON MARS TEST SET

Model	Mars-only	Mars+Earth	Change
Logistic Regression	0.813	0.826	+1.3%
Mini-CNN (10 epoch)	0.904	0.915	+1.1%
MobileNet (10 epoch)	0.940	0.916	**–2.4%**
Mini-CNN (15 epoch)	0.931	0.897	**–3.4%**
MobileNet (15 epoch)	0.896	0.878	**–1.8%**

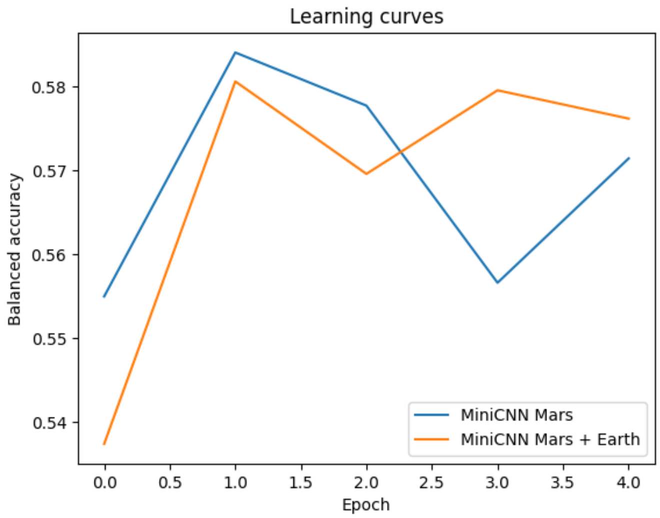

Fig. 9. Mini-CNN learning curves (balanced accuracy) on 15 epoch trial.

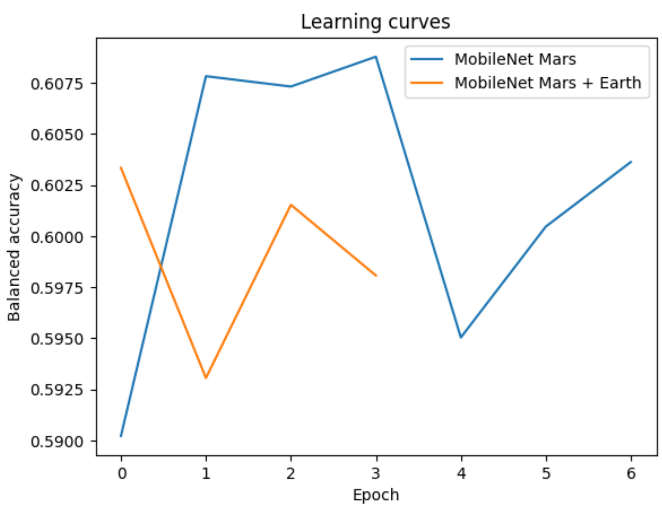

Fig. 10. MobileNetV2 learning curves (balanced accuracy) on 15 epoch trial.

Table II summarizes the balanced accuracies across all experiments. While co-training improved the baseline (logistic regression), it consistently reduced performance for MobileNet and led to instability in the Mini-CNN's deeper training run. This tells us that model complexity, training duration, and batch size have a significant impact on the effectiveness of domain co-training.

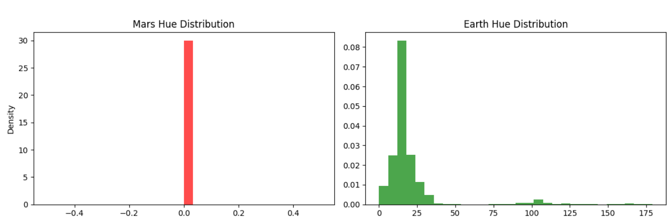

Fig. 11. Hue distribution of Mars vs. Earth training samples.

As shown in Figure 11, the Mars image data seems to have a narrow hue distribution (centered at 0), while the Earth images seem to span a broader range of hues. This difference is what introduces some domain shift and may explain why co-training reduces performance in deeper models (Table II).

V. Discussion

This study investigated the effectiveness of co-training with Earth and Mars data for terrain classification using lightweight convolutional neural networks. There were several key insights that emerged from the results.

A. Key Insights

Firstly, even a simple co-training method using Earth data (Freiburg Forest) alongside Mars data (AI4Mars) significantly improved model generalization. Across both the 10- and 15-epoch trials, Mars+Earth training improved recall for the minority *hazard* class. This supports earlier research that unseen domains (Earth-domain data in this case) can help models learn better, especially when positive hazard labels are sparse [11].

Secondly, different model architectures benefited to varying degrees from the co-training. The Mini-CNN showed a consistent gain in the 10-epoch trial, whereas the deeper MobileNetV2 model experienced performance drops in the Mars+Earth setting for both trials. These outcomes align with some known issues in heterogeneous-domain training, where complex models tend to overfit or misalign when domain shifts are not explicitly addressed [8], [11], [18].

Third, although MobileNetV2 achieved the highest balanced accuracy overall (94.0%), it did so using purely Mars data and did not benefit from the additional Earth data, whatsoever, in either trial. This is similar to a result found by Atha et al. [1], that pre-trained feature extractors can actually learn relatively well from low-variance imagery. In contrast, they may require some regularization or adaptation layers to avoid degradation when mixing domains.

B. Implications

For researchers, the results reinforce the need for domain-aware methods when introducing supplementary data. While datasets such as Freiburg Forest [20] may be visually similar to Martian terrain in some aspects, the low-level statistics differ quite a bit. Techniques such as domain-adversarial training [11], self-supervised continual learning [21], or zero-shot vision-language fusion [15] may offer more robust alternatives to naive co-training.

In terms of real-world application, the findings suggest that small, purpose-trained models like Mini-CNN can perform competitively, especially when carefully tuned and trained with domain-specific early stopping. Although MobileNetV2 [16] provides better peak accuracy on Mars data alone, it may not be the best candidate in resource-constrained missions where domain drift is expected. The selection of a model should balance accuracy, robustness to domain shift, and onboard resource limitations.

VI. Limitations

While the experimental findings provide us with valuable information about the impact of domain transfer on Martian terrain classification, the study also had several limitations, as listed:

- **Limited dataset diversity:** The Freiburg Forest dataset, although useful as a terrestrial representative, is not a perfect visual or semantic match for Martian terrain. Including it may have introduced biases or irrelevant features that, at the end of the day, affected model performance during co-training.
- **Lack of explicit domain adaptation:** This study explored naive co-training without any domain adaptation techniques. The resulting performance degradation in MobileNetV2 highlighted the need for more advanced methods such as feature alignment, domain-adversarial training, or style transfer.
- **Binary classification formulation:** By simplifying terrain classification to a binary problem (safe vs. hazardous), the study may have overlooked more diverse terrain features that could potentially impact real-world navigation. Using multiclass or continuous risk scoring, in the future, could offer more informative outputs that are relevant to rover applications.
- **Single test domain:** The evaluation phase was conducted exclusively using Martian data. While this was appropriate for the problem setting of this study, it also limits the generalizability of the findings to other environments or planetary surfaces.

VII. Conclusion

This project investigated whether co-training convolutional neural networks on both Martian and terrestrial datasets could improve binary terrain classification on Mars. Three models of varying complexity (logistic regression, a custom Mini-CNN, and MobileNetV2) were trained and tested under Mars-only and Mars+Earth configurations. The results showed that co-training provided modest improvements for simpler models, but degraded performance for deeper models such as MobileNetV2.

These findings indicate that domain transfer may be beneficial only when the target model is limited in capacity or trained on a small number of epochs. In contrast, high-complexity architectures like MobileNetV2 achieve better results when trained exclusively on domain-relevant data. This suggests that the effectiveness of domain transfer is context-dependent and sensitive to both dataset alignment and model complexity.

Future work could explore:

- Using domain adaptation techniques such as adversarial training or feature alignment to mitigate distribution mismatch between Earth and Mars data.
- Implementing self-supervised or online continual learning frameworks that can adapt during deployment in dynamic environments [21].
- Fine-grained or multi-label terrain classification to improve rover decision-making beyond binary traversability.
- Using cross-domain data augmentation techniques such as CycleGAN or style transfer to increase robustness without requiring manual labels.

The study ultimately provides a foundation for evaluating co-training strategies and highlights the importance of domain-aware learning in the field of robotic terrain perception.

VIII. Replication Package

- **GitHub Repository Link:** https://github.com/BrownAssassin/Big-Data-Analysis-Project
- **Publicly Accessible Overleaf Link:** https://www.overleaf.com/read/rzsgxgdykdtd#73f472

References

[1] Atha, D., Swan, R.M., Didier, A., Hasnain, Z. & Ono, M. 2022, 'Multi-mission Terrain Classifier for Safe Rover Navigation and Automated Science', *IEEE Aerospace Conference*, pp. 1-13, DOI:10.1109/AERO53065.2022.9843615.

[2] Buslaev, A., Parinov, A., Khvedchenya, E., Iglovikov, V. Kalinin, A. 2025, *Albumentations: fast and flexible image augmentations (version 1.3.1, software)*, documentation, viewed 19 August 2025, <https://albumentations.ai/docs/>.

[3] Frey, J., Patel, M., Atha, D., Nubert, J., Fan, D., Agha, A., Padgett, C., Spieler, P., Hutter, M. & Khattak, S. 2024, *RoadRunner: Learning traversability estimation for autonomous off-road driving*, arXiv preprint, arXiv:2402.19341v3, viewed 19 August 2025, <https://arxiv.org/abs/2402.19341v3>.

[4] Hirose, N., Sadeghian, A., Vázquez, M., Goebel, P. & Savarese, S. 2018, *GONet: A semi-supervised deep learning approach for traversability estimation*, arXiv preprint, arXiv:1803.03254, viewed 19 August 2025, <https://arxiv.org/abs/1803.03254>.

[5] Jeon, Y., Son, E.I. & Seo, S.-W. 2024, *Follow the Footprints: Self-supervised Traversability Estimation for Off-road Vehicle Navigation based on Geometric and Visual Cues*, arXiv preprint, arXiv:2402.15363v1, viewed 19 August 2025, <https://arxiv.org/abs/2402.15363>.

[6] Jiang, P., Osteen, P., Wigness, M. & Saripalli, S. 2020, *RELLIS-3D Dataset: Data, Benchmarks and Analysis*, arXiv preprint, arXiv:2011.12954v4, viewed 19 August 2025, <https://arxiv.org/abs/2011.12954v4>.

[7] Mattamala, M., Frey, J., Libera, P., Chebrolu, N., Martius, G., Cadena, C., Hutter, M. & Fallon, M. 2024, *Wild Visual Navigation: Fast Traversability Learning via Pre-Trained Models and Online Self-Supervision*, arXiv preprint, arXiv:2404.07110v1, viewed 19 August 2024, <https://arxiv.org/abs/2404.07110>.

[8] Mori, J., Furukawa, R., Teranishi, I. & Sakuma, J. 2023, *Heterogeneous domain adaptation with positive and unlabeled data*, arXiv preprint, arXiv:2304.07955v2, viewed 19 August 2025, <https://arxiv.org/abs/2304.07955v2>.

[9] NumPy Developers 2025, *NumPy: the fundamental package for scientific computing with Python (version 1.26, software)*, documentation, viewed 19 August 2025, <https://numpy.org/doc/>.

[10] OpenCV Project 2025, *Open Source Computer Vision Library (version 4.x, software)*, documentation, viewed 19 August 2025, <https://docs.opencv.org/4.x/index.html>.

[11] Palazzo, S., Guastella, D.C., Cantelli, L., Spadaro, P., Rundo, F., Muscato, G., Giordano, D. & Spampinato, C. 2020, *Domain adaptation for outdoor robot traversability estimation from RGB data with safety-preserving loss*, arXiv preprint, arXiv:2009.07565, viewed 19 August 2025, <https://arxiv.org/abs/2009.07565>.

[12] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M. Duchesnay, E. 2025, *scikit-learn: machine learning in Python (version 1.4, software)*, documentation, viewed 19 August 2025, <https://scikit-learn.org/stable/>.

[13] Peng, J., Estrada, G., Pedersoli, M. & Desrosiers, C. 2020, 'Deep co-training for semi-supervised image segmentation', *Pattern Recognition*, vol. 107, DOI:10.1016/j.patcog.2020.107269.

[14] PyTorch Contributors 2025, *PyTorch documentation (main branch)*, software documentation, viewed 19 August 2025, <https://docs.pytorch.org/docs/main/>.

[15] Sahu, S., Singh, A., Nambiar, K., Saripalli, S. & Sujit, P.B. 2025, *AnyTraverse: An off-road traversability framework with VLM and human operator in the loop*, arXiv preprint, arXiv:2506.16826v1, viewed 19 August 2025, <https://arxiv.org/abs/2506.16826v1>.

[16] Sandler, M., Howard, A., Zhu, M., Zhmoginov, A. & Chen, L.-C. 2019, *MobileNetV2: Inverted Residuals and Linear Bottlenecks*, arXiv preprint, arXiv:1801.04381v4, viewed 19 August 2025, <https://arxiv.org/abs/1801.04381v4>.

[17] Seo, J., Kim, T., Kwak, K., Min, J. & Shim, I. 2023, *ScaTE: A scalable framework for self-supervised traversability estimation in unstructured environments*, arXiv preprint, arXiv:2209.06522v2, viewed 19 August 2025, <https://arxiv.org/abs/2209.06522v2>.

[18] Sevastopoulos, C. & Konstantopoulos, S. 2022, *A survey of traversability estimation for mobile robots*, arXiv preprint, arXiv:2204.10883, viewed 19 August 2025, <https://arxiv.org/abs/2204.10883>.

[19] Swan, R.M., Atha, D., Leopold, H.A., Gildner, M., Oij, S., Chiu, C. & Ono, M. 2021, 'AI4Mars: A dataset for terrain-aware autonomous driving on Mars', *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pp. 1982–1991, DOI:10.1109/CVPRW53098.2021.00226.

[20] Valada, A., Oliveira, G., Brox, T. & Burgard, W. 2016, *Deep Multispectral Semantic Scene Understanding of Forested Environments using Multimodal Fusion*, International Symposium on Experimental Robotics, viewed 19 August 2025, <https://deepscene.cs.uni-freiburg.de/>.

[21] Yoon, H.-S., Hwang, J.-H., Kim, C., Son, E.I., Yoo, S.-W. & Seo, S.-W. 2024, 'Adaptive robot traversability estimation based on self-supervised online continual learning in unstructured environments', *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 4902–4909, DOI:10.1109/LRA.2024.3386451.